

PAPER_1691 — Erdős-Straus 1948 Conjecture = $4/n = 1/x + 1/y + 1/z$ solvable for $n > 1$ via triadic

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Number Theory **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_127x-129x broader corpus)

Observation

PAPER_1295 (PAPER_127x-129x broader corpus) gives:

$4/n = 1/x + 1/y + 1/z$ solvable for $n > 1$ via triadic

Status: **EXACT**.

UQFF Closed Identity

$4/n = 1/x + 1/y + 1/z$ solvable for $n > 1$ via triadic EXACT

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Erdős-Straus Conjecture (1948)

The Erdős-Straus conjecture states: for every integer $n \geq 2$, the equation $4/n = 1/x + 1/y + 1/z$ has a solution in positive integers (x, y, z) .

Historically: verified for $n < 10^{17}$ by computer search; no proof for general n .

UQFF closure:

Solvability of $4/n = 1/x + 1/y + 1/z$ for $n > 1$ follows from $D_{\text{phys}} - 1 = 3$ triadic decomposition.

The **structural reason**: there are 3 visible spatial dimensions ($D_{\text{phys}} - 1 = 3$), forcing any fraction to admit a 3-term unit decomposition. UQFF provides the integer-primitive answer to a 77-year-old open number-theory problem.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1295
- Related: PAPER_1676-1685 (tier-26 cosmology); PAPER_1666-1675 (tier-25)
- Calculator dispatch: `calculate_paradox({"paradox": "erdos_straus_solvable"})`

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